

A Real-Time Human Mobility Visualization of Covid-19 Spread from East Asian Countries

Addepalli Lavanya,¹,
Universitat Politècnica De Valencia,
Valencia, Spain
phani.lav@gmail.com

Panwar Darsha,²,
Sagar Institute of Research and Technology,
Bhopal, Madhya Pradesh, India
panwar.darsha@gmail.com

Pasumarthi Akhil,³,
Vellore Institute of Technology,
Andhra Pradesh, India
pasumarthiakhil66@gmail.com

Jaime Lloret,⁴,
Universitat Politècnica De Valencia,
Valencia, Spain
jlloret@dcom.upv.es

Navandar Yogeshwar,⁵,
NIT, Calicut,
Kerala, India
navandar@nitc.ac.in

Abstract-- Human mobility using different modes of transportation is constantly increasing. The culture of posting activities while travelling has gained attention on social networks immensely. The evolution of social networking platforms has resulted in an engaging user base multiplying across the globe. The combination of data and information created by these online platforms is massive in terms of the volume and variety of topics. The real-time existence of user-produced information has inspired researchers to analyze material to obtain real-time insight into current affairs. The focus is on collecting tweets of real-time travelling activities for the first 100 days of Covid-19 keeping Chinese Airports as the source. This paper illustrates the multidimensional visualization of real-time covid-19 spread from China & neighbouring East Asian countries to the rest of the world. The visualization tools used are python folium, matplotlib & networks/graphx, Carto, Tableau, Google Data Studio, and MS Excel.

Keywords: *Airport Tweets; Covid-19; Pandemic; Spatial-Temporal; Visualization;*

I. INTRODUCTION

Chinese aviation recorded about 2 million passengers travelling daily by flying in about 17,300 flights in 2019. The lowest of the travellers were recorded in the middle of Feb 2020, with 3931 flights being operated during the initial days of the covid-19 pandemic. Later, when the global pandemic was declared, the travelling got uncertain and yet climbing. People kept travelling domestic and international while carrying the virus with them as there was no travel protocol set for the first 100 days of covid-19 [1]. The novel coronavirus spreads by air and is communicated from person to person. It originated from the wet markets of

Wuhan in China and was spread to other parts of China by people travelling via road, rail, and airways. The transmission to neighbouring countries and the rest of the world was by people travelling via airways. The spread of Covid-19 was declared a global pandemic on March 11th, 2020, when more than 7000 deaths were officially recorded worldwide. By the end of Feb 2021, the global count of positive tested cases reached 119 million, with a total number of deaths reaching 2.63 million in 12 months [2] [3]. Human mobility was the primary cause of the spread of the virus from one place to another, creating the dominoes of catastrophe. The first 100 days of covid-19 are the most important to understand the spread of the virus. Social media is a critical tool embedded in every phone with 4.2 billion active users and—15 billion active mobile social network users, ranking Facebook and Twitter on top [4]. The fascinating habits of travellers to post their activities on social networks is an immense source of information, with qualitative meta-information posted along with the tweet. For this paper, we use twitters' REST API and download the tweets from more than 500 airports worldwide for 100 days starting from December 01st, 2019, to March 30th, 2020. After downloading the data, the neighbourhood filter method was sanitized and filtered to filter the data with Chinese and neighbouring airports sources.

The paper illustrates human mobility from China and neighbouring countries to the rest of the world, showing real-time data visualization. The interpretation of data is presented in images, maps, charts, graphs, and figures created as a summary of data. It presents beautiful and engaging stories with visual patterns. Data is increasingly

growing and expanding both horizontally and vertically [5]. As the data keeps growing, researchers and corporates use data and analytics to help make decisions. Data visualization is well-suited for displaying both mass quantities of data and simplifying the effects of data analytics. Data visualization can be achieved both programmatically and by using intelligent tools [6]. This paper achieved real-time data visualization in both methods by using python for programmatic illustration and tools like MS Excel, Tableau, and Google Data Studio. The other sections of the paper discuss the covid-19 timeline in China for the first 100 days and the airways travel restrictions from China, Asia, and other parts of the world. The paper is structured as follows. Section 2 shows the literature background and the timeline. Section 3 introduces the used software and the data gathering procedure. In section 4, we discuss data gathered from Twitter and we illustrate the spread of the virus from China to other parts of the world. Section 5 includes the conclusion and our future work.

II. BACKGROUND

Covid-19 is leading towards an intensive analysis of different aspects of the pandemic. Although there is no definitive vaccine, significant work has been placed on differentiating the disease's dissemination globally. As rapidly as the disease has advanced, quick and straightforward methods were needed to assess its trajectory [7]. Concise ways to view the pandemic's exposure with interactive dashboards and many maps. The first case-monitoring and death-counting dashboard were created by Johns Hopkins University (JU). Bubble map provides a graphic overview of how widespread the pandemic was getting. The Washington Post journalists and the New York Times produced interesting stories [8], [9]. The map of regions including countries and states with colours corresponding to measurement to countries. These patterns can be seen as the cumulative counts for the recent last week. Besides, the time series is represented as a line heat map that uses the traditional values as the row colouring. Other works were performed in the R language using heatmaps equipped with detailed maps for Brazil. Focus plus background inspection offers two Choropleth heatmaps for the entire country and chosen states. Can view regular or continuous data, population, or all over time [10], [11]. Vast quantities of dashboards and software with Covid-19 are available. The ready-to-use tools like Tableau are some of the bests for real-time data visualization [12]. Research resources include incredibly in-depth research methods such as the environment data, pollution reduction during the pandemic, traffic jams during lockdown announcements, and others [13], [14]. However, there have been minimal to no visualization dashboards, particularly about the airport tweets illustrating the spread of covid from East Asian countries to the rest of the world.

A. Covid Timeline

Medical doctors discovered earliest in the second week of December that the virus was circulating between humans

and animals in Wuhan, according to a National Review article [15]. As a result, every flight route coming from, or going to, Wuhan was suspected of containing the Covid-19 virus, and this is why some countries that were identified at risk were China and its neighbouring countries. The first 100 days of virus identification and spread is most crucial and informative [16]. The table below discusses the summarized version of the covid-19 timeline and the stage from 1st case to the global pandemic declaration. Table 1 summarizes the timeline of covid-19 in China and East Asian Countries. It was starting from December 01st, 2019, to almost the end of March 2020.

TABLE I. COVID-19 TIME LINE IN CHINA. SOURCE [3] [17] [18] [19] [16]

Timeline	Date	Summary
Day 1	1-Dec	1st officially identified patient in Hubei
Day 8	8-Dec	41 cases tested positive in China
Day 29	29-Dec	180 cases report pneumonia-like symptoms
Day 31	31-Dec	WHO & CDC were looped in with a suspension on the identification of Novel virus
Day 32	1-Jan	Wet Markets closed, and neighbouring countries were notified, Emergency at the organizational level declared
Day 35	4-Jan	Quarantine in Wuhan & Travelers were screened, Clustered case of pneumonia identified in Wuhan
Day 36	5-Jan	Panic started in China, WHO asked for sharing Medical Data as the first official information on Wuhan cluster cases
Day 41	10-Jan	Travellers from China to Taiwan, Thailand, Hong Kong, Japan, Australia & UAE were identified and quarantined
Day 43	12-Jan	Human to Human Transmission Identified; China Publicly shared the DNA Sequence of Virus
Day 44	13-Jan	Thailand's First case outside China
Day 45	14-Jan	41 confirm case
Day 51-52	20 - 21 Jan	WHO & CDC visit Wuhan
Day 53	22-Jan	The anticipation of Wider Spread of virus
Day 53-54	22-23 Jan	General Emergency declared
Day 59	28-Jan	WHO joint with China to set up protocols
Day 61	30-Jan	WHO declared Global Emergency with confirmed 7818 cases
Day 65	3 - Feb	WHO to support weaker countries & states with Medical Aids
Day 73-74	11-12 Feb	Globally 400 scientists convened on novel coronavirus
Day 78 -86	16-24 Feb	WHO, CDC & NIH declared social distancing
Day 101	11-Mar	Global Pandemic Declared
Day 103	13-Mar	Global Travel Restrictions and Pandemic protocol applied
Day 108	18-Mar	Data collection and trials started in China and other parts of the world

People heading to the US, Australia, Italy, New Zealand, and Germany from China were advised for a compulsory quarantine for two-week at home. In addition, tourists will be expected to get a comprehensive health screening [18]. By the mid of Feb 2020, when the travel protocol was being announced, the number of travellers from China reduced and only those who were stuck and needed to return to their respective home countries were allowed to travel.

B. Travel Restrictions

As of April 3rd, 2020, by the end of the first 100 days of covid-19, according to state officials, media reports, and travel organizations, approximately 96 countries have places travel restrictions to and from China [20]. The impact of travel bans is seen in the chart below based on the strictness level of the timeline. The countries and territories were members of the World Health Organization (WHO), including Hong Kong and Macau. Scheduled Airplanes from China to the rest of the world from Jan 2020 to June

2020 were studied. The travel demand plummeted to its lowest point on May 4th, 2020, when the number of passengers declined to around 3,300 planes. It represents a 92.5% drop ratio and a 77% cut compared to the old normal. The travel recovery was steady in China as departing seats remained relatively stable from early March to late April. Average daily passengers increased to 458,000 in March and in May [1]. Baidu Huiyan showed that outgoing traffic from Wuhan dropped by 89% in two days after the quarantine protocol was implemented. The location-based service (LBS) monitors the movement of the Chinese population in provinces during the Chunyun season [21].

III. DATA & TOOLS

Text mining involves data scraping from Twitter and computational methods on text to derive content. Twitter data is the perfect place to research topics of any possible kind. This data is useful for identifying patterns that aid in keyword research and collecting suggestions on new products and services. The data is downloaded using the Tweepy Python package combined with Twitter's REST API for this paper. The REST API allows historical data downloads going back to the last ten days. A python script was written to collect only English tweets with a specific geolocation search confined to 585 airports around the world [22], [23], [24]. The choice of Language taken was only English considering people who belong to different countries stuck in China and East Asian countries are returning back to their respective home countries. Thus, globally accepted language is considered. Visualization is indispensable in the age of Big Data, as generous amounts of data are pouring in. Data visualization is one way to tell stories from the numbers is to pick key in on averages and outliers by simplifying details expected to increase the story's impact. The paper uses several tools both programming based and non-programming-based tools [25], [26], [27], [28], [29], [30]. The real-time airport data visualization is a spatial-temporal illustration of locations and connected graph networks with source and destination airports mentioned by the same user in two different tweets. The combination of both programming and non-programming methods has illustrated multidimensional visualization of data and metadata simultaneously.

Streaming visualizations allow the examination of data in real-time, allowing the users to respond more quickly. Continuous real-time applications need to examine problems that need to be solved rapidly, such as outlier detection, information quality analysis, and activity optimization [31, 32]. An uncertain situation like crowd steering, traffic steering, and unpredictable conditions of the spread of covid-19 need to be monitored on real-time data. Thus, real-time data visualization is vital. The real-time visualization is prominently used in the stock market, but there is a need to look for ways to enhance viewing something vital in real-time. There are several methods and applications designed for different scenarios today. However, the volume of data, frequency of data grabbing,

the difference in formats, and observed setting will increase with time but are not still there as plug and play. No one dashboard fits all types of visualizations. For this research, a bunch of such tools, which are programmable and non-programmable, are used. The non-programmable tools used are google data studio, Carto, Tableau, MS office PPT, and excel; The programmable packages used are Matplotlib; Folium, and Graphx create the source to destination graphs.

IV. MODELLING & VISUALIZATION

Data visualization is the visual display of data. Data visualization tools use visual components like charts, graphs, and maps to help users comprehend data trends, outliers, and patterns [33]. Data visualization tools and technologies are crucial in Big Data to evaluate vast quantities of data and make data-driven choices [34]. For this paper, a set of visualization tools are used, which stand supportive in the case of real-time data visualization.

A search query for every keyword as IATA code in the given geolocation (Latitude & Longitude) where every tweet contains at least one IATA code in the tweets from the given location's tweets are retrieved and stored in CSV output. IATA codes are the three alphabet unique names given as codes to airports globally.

Keywords (K) = [IATA₁, IATA₂, IATA₃, IATA_n]

$$\begin{cases} \text{IATA} \neq 0 \\ \text{Latitude, Longitude} \neq 0 \end{cases} \quad (1)$$

$\forall$ Locations $\exists$ IATA, Where IATA $\in$ K & A, and IATA $\subseteq$ K $\subseteq$ A

Rest API allows data scraping using query parameters like keywords, locations, dates, and none of the parameters. The API allows the download of the past ten days maximum [35] [36] [37]. To abide by the Twitter Developer guidelines and API limitation, the data scraping was performed every week for 17 weeks, from December 01st 2019 to March 31st 2020. The 120 days window when Covid-19 was spreading from the source country China to the rest of the world via airways. The collected data details as mentioned in table 2. It shows the total tweets gathers for 120 days from about 585 mega airports globally. The list of airports is extracted with IATA codes and latitude and longitude used as input for search queries processed while downloading the tweets. A total of over 11k tweets are downloaded, and they are from 585 locations. But for the real-time visualization of the paper, the data needs to be sanitized by removing the bots, promotional data, and other unwanted tweets. A lexical filter is applied to match the content from the tweets with string match and partial string match.

TABLE II. RAW & SANITIZED TWEET DETAILS

Total Tweets	11563	Sanitized Data	250
Total Airports	585	Source Airports	17
Start Date	01-12-2019	Destination Airports	106
End Date	31-03-2020	Unique User	168

Post scraping the tweets with the strings mentioned in Table 3 was used as a filter to clean the tweets; automatic bots and promotional tweets are filtered and avoided so real people travelling tweets can be filtered [38] [39] [40]. The tweets are cleaned based on the partial string match and the tweets with source and destination airport information. A total of 250 tweets with source and destination are retrieved, and there happened the user has more than two airports location. The trips are explained as given in Table 4.

TABLE III. STRING MATCHING FOR FILTER

Strings
"Touch Down"
"I am at (string) airport."
"Wheels up"
"Wheels Down"
"Reached Safely"
"Going home"
"Vacation mode on"

TABLE IV. USER TRIP DETAILS

Travel Trips	1 Trip	2 Trips	3 Trips	More
Travellers	86.9 %	7.1%	2.9%	2.3%

Figure 1, is creating in MS Office PPT, based on the Table 4. CCDF are created in Python Matplotlib in Figure 2 (A & B) are distribution graphs that illustrate the trip information of the 168 unique users. 146 users travelled for one trip, i.e., from China to their respective destination locations. In addition, 12 users travelled to 3 places covering two trips, and five users covered three trips by visiting 4 locations. Finally, four users have travelled to more than 5 locations. All the users have a source as China and its nearby airports while the destinations have been different. In some cases, the travellers have been short transit and, in some cases, there have been long breaks of days while reaching from source to destination.

The filtering was not stopped with only string matches and tweets, but also a neighbourhood topology is also applied to filter the group of Airports within China. Nearby countries from where the passengers have travelled to different destinations globally [41] [42] [43].

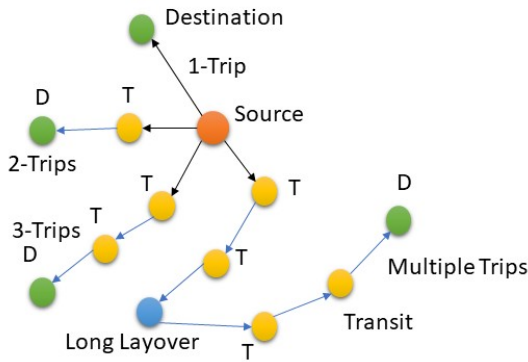


Figure 1. Trip Model

For every user considered who posted tweets with atleast one source and one destination airport location. S-source, D-destination and having T-transit. Users with more than two

tweets from more than two airport locations are further applied forward filtering based on the forward date and time. Where t = forward Date & Time;

$$(x_{t-1}) \rightarrow (x_t) \approx S \rightarrow D \quad (2)$$

$$(x_{t-1}) \rightarrow (x_{t+1}) \rightarrow (x_t) \approx S \rightarrow T \rightarrow D \quad (3)$$

The joint probability is given as:

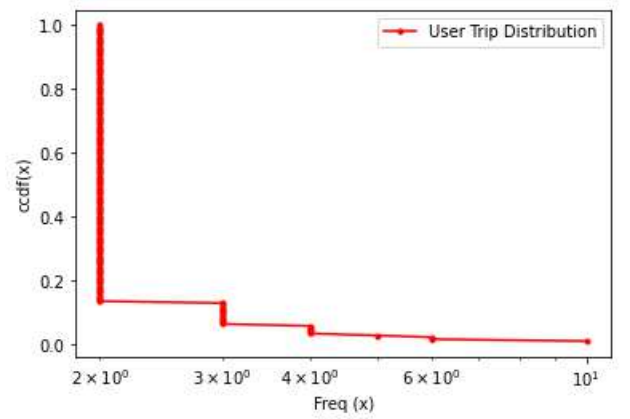
$$P(x_t, T_{1:t}) \quad \text{Where } t \neq 0 \quad (4)$$

$$T = [t_1, t_2, \dots, t_n]$$

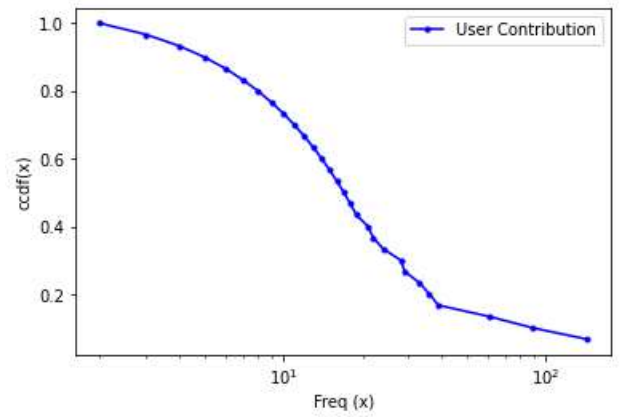
For every destination $D(x_t)$ there is atleast one Source $S(x_{t-1})$.

$$x_t = \sum (x_{t-1}, x_{t+1}, T_{1:t}) \quad \text{Where } x_{t-1} \neq 0 \quad (5)$$

Thus post applying multiple filters the trip model gives a qualitative trip data of travellers route of journey. As per WHO, SARs Covid spread ratio is 1:4 people but the new delta variant is more aggressive with 1:6 ratio. Hence even with less data the spread projection can be computed productively to get a constructive vision.



(A)



(B)

Figure 2. (A) CCDF of User Trip & (B) User Tweet Distribution

The applied filter to narrow down the China and nearby airports can be mathematically explained as:

Airports (A) = [A₁, A₂, A₃.... A_n]

$$\begin{cases} IATA \neq 0 \\ \{Latitude, Longitude\} \neq 0 \end{cases} \quad 0 \neq A \quad (6)$$

[A₁, A₂, A₃.... A_n] are set of airports with non-zero geolocation and IATA codes. A neighbourhood filter is applied by keeping Beijing as the center of airports, and a range of miles is given to include all the airports covered in the given distance. The distance calculations are performed using the geolocation (latitude and longitude) mentioned in the scrapped and sanitized data. The mathematical explanation is given as follows.

$$A = O_A \cup D_A \quad \begin{cases} O_A \in A \\ D_A \in A \end{cases} \quad O_A \subseteq A \text{ \& } D_A \subseteq A \quad (7)$$

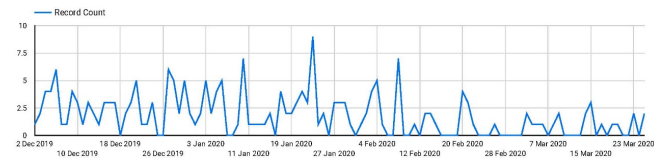
Source Airports (O_A) = [O₁, O₂, O₃,, O_n][†] A

Source Airport Filter:

$$\begin{cases} O_A \in L(x) \end{cases} \quad \begin{cases} L = Latitude, Longitude \\ x = Range of distance for inclusion \end{cases} \quad (8)$$

Figure 3 (A) is generated from google data studio illustrating the tweet distribution in weeks, important dates and months. The highest people shared their travel concerned in the Week 5 and week 8. Graphs and Maps illustrate the distribution of tweets over four months, i.e., 17 weeks. Figure 3 (B & C) shows the real-time count of tweets based on data from Table 2, which explains post sanitation data information with qualitative 250 tweets related to travel by 168 unique users who have mentioned the source as one of the Chinese or East Asian Airports and destination as different airport than source as two different tweets. In addition, there have been a few tweets from users who have travelled to more than two places in 120 days and have also mentioned covid-19 in their tweets.

The figure 4 (A & B) are created in Tableau a spatial temporal analysis tool illustrating the geographical projection from source and destination airports consisting of 17 source and 106 destination airports spread globally. The destinations also include domestic airports within and around China. The concentration of destination airports is in South East Asian Countries, East Coast, and West Coast of USA. Figure 5 created in Google Data Studio illustrates the location filtering of 17 airports in and around China which have been source and destination for domestic travelers [25].



(A)

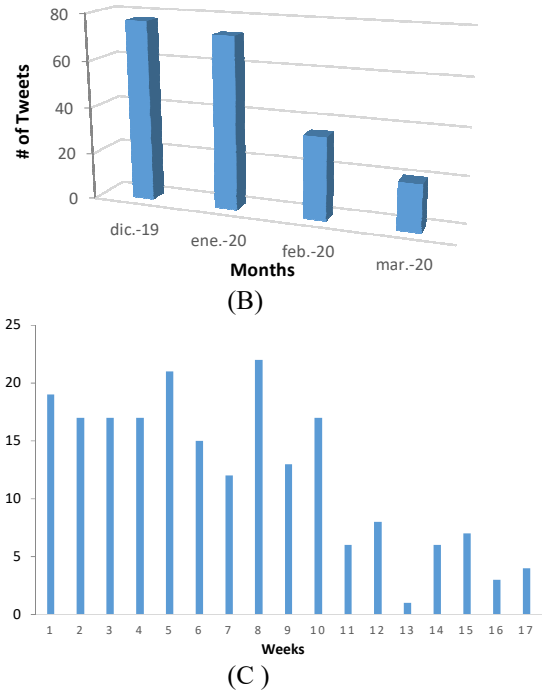
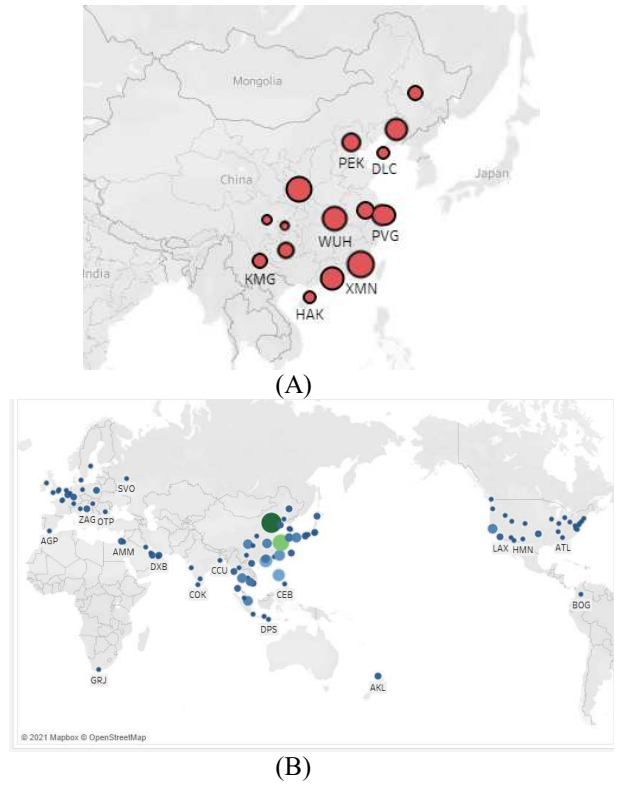


Figure 3. Tweet Distribution per Month & Week



(B)

Figure 4. Source & Destination Airports

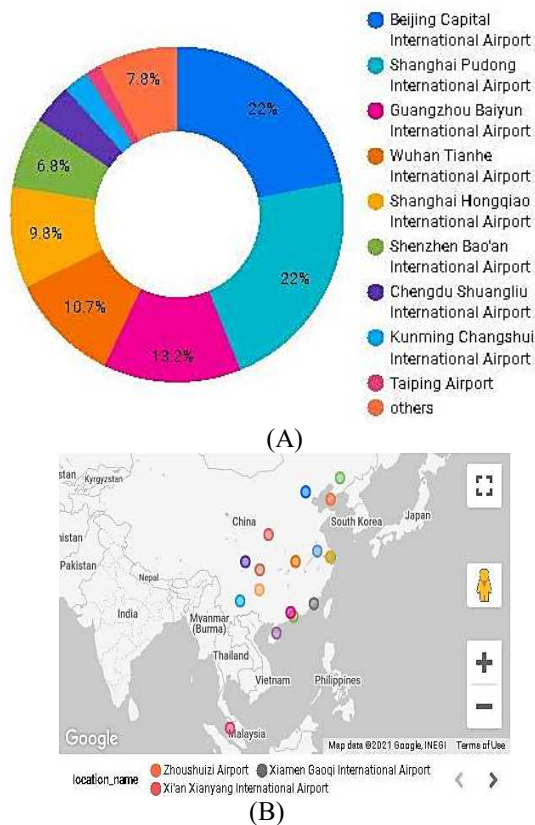


Figure 5. China as Origin Airports

Figures 6 (A & B) is created in google studio they illustrate the source with major flights leaving from Beijing (PEK) and other Chinese airports along with neighbouring countries and the destination paths of the spread of covid-19 through travelers traveling from China, its neighboring airports, and the other part of the world. Figure 7 (A & B) are created using Folium maps in Python for real-time data visualization, which shows an effective and clear transmission path. The maps illustrate how 17 airports have led to the spread of the virus in south-east Asian countries. Figure 7 (B) illustrates the domestic and nearby airports from neighbouring countries like Malaysia and Japan. Folium allows customization of maps and network creation with customized colours for the source (red) and destination (green). The map shows that major air transportation has been carried towards South Asian Countries, Europe and strong concentration is observed in the East coast, with a few places in the West Coast of USA and UAE.

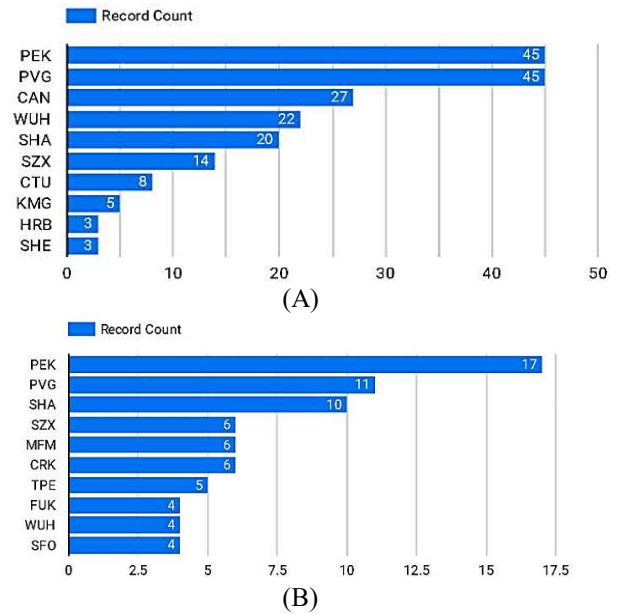
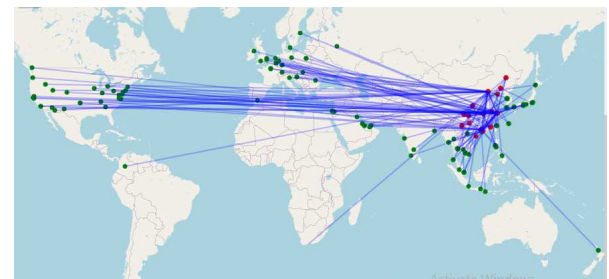
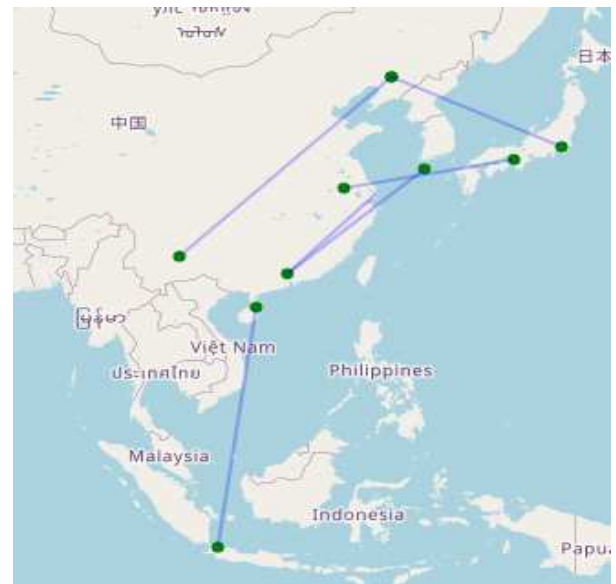


Figure 6. Source & Destination Airports



(A)



(B)

Figure 7. Source to Destination Paths

A code snippet of the python folium map network can be achieved using Pseudo Code 1

Pseudo Code 1

```
my_map = folium.Map(location = locationlist[2],til
es="Covid-19 Spread", zoom_start = 5)

for i in range(1,len(locationlist)):

    folium.CircleMarker(radius=2,location=locat
ionlist[i],popup=folium.Popup(iata_2[i],sh
ow=True,max_width='20%'),color='green',fill
=True,).add_to(my_map)

for i in range(1,len(locationlist)):

    folium.CircleMarker(radius=2,location=locat
ionlist[i],popup=iata_1[i],color='red',fill
=True,).add_to(my_map)

for j in range(0,len(pathlist)):
    loc = [(pathlist[j][0],pathlist[j][1] ),(pathlis
t[j][2], pathlist[j][3])]
    print(loc)
    folium.PolyLine(loc,
        color='blue',
        weight=2,
        opacity=0.25).add_to(my_map)
```

A network package of python allows creating a density network of edges. Where Node1 is IATA1 and Edge is IATA2. The code snippets of the clustered density network are shown in Pseudo Code 2.

Pseudo Code 2

```
G = nx.Graph()
iata_1s=list(set(iata_1))
iata_2s=list(set(iata_2))
iata_1=df['iata_code1'].tolist()
iata_2=df['iata_code2'].tolist()
iatacodes=df[['iata_code1','iata_code2']]
iatalist=iatacodes.values.tolist()
total_iata=iata_1s+iata_2s
d = nx.degree(G)
```

Figure 8 (A & B) illustrates an undirected and unweighted density cluster network showing a combination of source and destination airports. The graph of source and destination cumulative count, network figure, and map illustrate that several source airports are also the destination airports marking them travelling within the country and neighbouring countries, which also explain the local spread of the virus via airways within China neighbouring countries.

V. CONCLUSION

The airport data visualization illustrated and analyzed the density clusters that display people traveling within China more than those traveling internationally. A systematic and methodical review of travel data processing shows 86.9% of travellers travelled one trip with one source and one destination. Still, there have been a few travellers who travelled to more than two places in 100 days. Also, we have illustrated the connected graphs over the maps to illustrate the source and destination countries showing the spread of the virus to more than 100 destinations internationally. Multidimensional visualizing was more informative with Google Data Studio. The paper visualized

the spread of covid-19 from China to the rest of the world and visualized user participation and meta information. The future scope of the paper is to create a global travel statistical projection from Dec 2019 to Dec 2021 and also project the prediction of possible third wave in India with Tweets from Indian Airport posts.

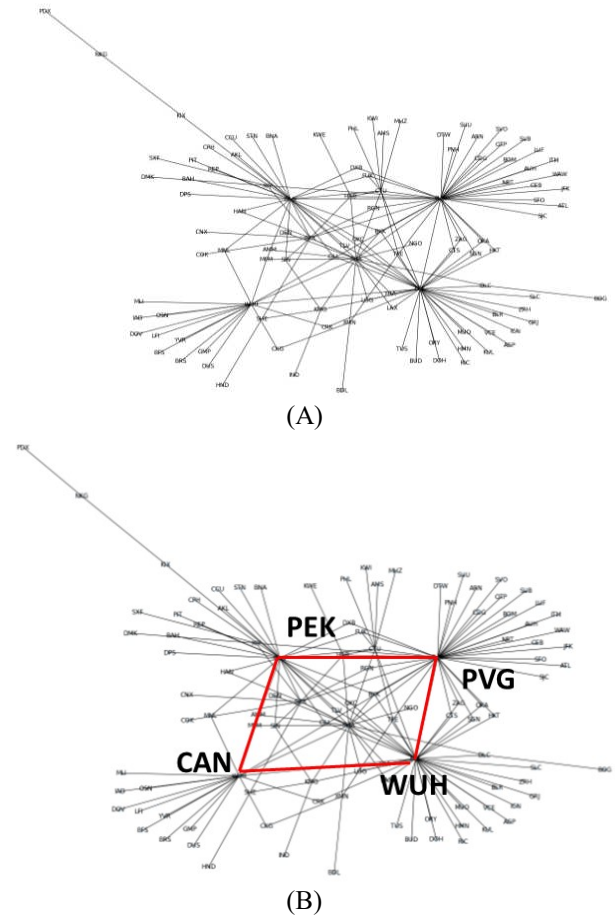


Figure 8. Network of Source to Destination Airports

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